Examination of the impact of range, cage-free, modified systems, and conventional cage environments on the labor inputs committed to bird care for three brown egg layer strains

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Primary Audience: Egg Producers, Complex Managers, Flock Supervisors, Researchers

SUMMARY

Although the poultry industry has developed into an intensive management system over the past several decades, current trends are moving toward extensive systems with decreased hen-stocking density. However, there is limited information on the resource demands of such systems, particularly with regards to labor requirements. Labor h commitment per hen originally housed and hens surviving were evaluated for 4 different environments: range, cage-free, modified cage, and conventional cage. Within each system, 3 strains of brown egg layers were examined to determine whether production system influenced labor h. Data collection began at 33 wk of age and continued until 89 wk in all environments. Range systems demonstrated the highest labor h requirement for both the hens originally housed and hens surviving measurements, particularly during the summer months when pasture management within the paddocks was time consuming. Conventional and modified cage systems required the least time commitment with cage-free serving as an intermediate. Cage-free labor h increased toward the end of the cycle as maintaining litter quality within the house became more demanding. The cost of labor h was not offset by the price per dozen eggs produced, and the difference was greater in the extensive systems. This study supports previous findings that extensive layer production systems require greater labor h and provides an economic perspective to the increased time commitment.

Key words: labor, cage, cage-free, free-range, economic cost

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DESCRIPTION OF PROBLEM

Over the past decade, the use of alternative production systems has increased in the commercial table egg industry. These changes have largely been driven by increased pressure from consumers arising from their perception regarding hen welfare within caged housing environments compared with alternative systems that provide increased space per hen (Anderson, 2009a; Kidd and Anderson, 2019). However, there is a paucity of research concerning how implementing alternative systems in place of

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conventional systems will impact hen welfare, the table egg market, or the producers. Of particular concern for egg producers is the extensive time commitment required to manage alternative systems in terms of labor h per hen or egg produced.

Much of the progress made in the poultry industry over the last few decades has been a result of improvements in bird genetics, nutritional advancements, and increased labor efficiency. Historical labor inputs for egg production ranged from 2.2 to 2.6 man hour per hen housed (**h/hh**), although this varied depending on the season and enclosure size (Lee, 1948; Winter and Funk, 1949; Juhl, 1951). The development of environmentally controlled, confined housing systems together with genetic selection of hens suited for intensive production environments has dramatically decreased the labor investment required to produce table eggs. Work comparing cage and range environments using modern breeds has demonstrated that the labor h/hh required for caged hens was significantly reduced to 0.173 h/hh, whereas that for range facilities remained similar to historic levels at 2.663 h/hh (Anderson, 2009b). More recent work at the same facility has demonstrated an improved efficiency in range management that reduced labor to 1.268 h/hh, however it remained the most labor intensive system, requiring approximately 3 times the labor h compared with the conventional cage system (Anderson, 2014). If larger commercial producers are to transition to cage-free and range systems in an economical manner, more information on the labor required to manage these systems is necessary. Given the improved production potential of modern commercial strains compared with heritage breeds, it is likely that they would be the preferred birds selected; however, their increased production may also require more labor h within the alternative systems and may actually be more costly than other breeds in terms of labor h/hh. Therefore, the objective of this study was to determine the man h needed to manage 3 commercial brown egg layer strains through a single production cycle. The economic implications of the differing labor costs required compared with eggs produced in each of these management systems were also examined.

MATERIALS AND METHODS

Hens from 3 commercial brown egg layer strains were housed within 4 different environments from 33 to 89 wk of age and the labor h required to manage these hens in different housing systems was recorded (Anderson, 2018). The birds were hatched and grown at the North Carolina Department of Agriculture's Piedmont Research Station in accordance with the methods used for the fortieth North Carolina Layer Performance and Management Test flock (Anderson, 2018). All procedures within this trial were reviewed and approved by the North Carolina State University Institutional Animal Care and Use Committee. Bird management and housing environments were also compliant with the housing standards of the United Egg Producers (UEP, 2017a,b) and the regulations set forth by the Food and Drug Administration Egg Safety Rule (USFDA, 2009). Hen numbers per replicate and housing system are provided in Table 1. Pullets assigned to the conventional and modified systems were reared in a similar caged environment before being transferred to their respective laying houses at 16 wk of age. Pullets assigned to the free-range and cage-free systems were reared within their respective laying facilities to encourage their learning to navigate these environments. The chicks were hatched concurrently and identified by a permanent identification tag identifying their assigned production system, as well as their replicate cage and bird number. All birds were maintained on the same dietary regimen, vaccination schedule, and lighting program, with the exception of the free-range birds that moved to natural light supplemented with controlled day lengths at 12 wk. Temperature profiles were similarly managed between all systems, with the exception of free-range as complete temperature control was not possible in the open-sided huts. A heater was provided within each hut so that the interior temperature was maintained at a minimum of 7.2°C (45°F). Feed and water were provided ad libitum throughout the experiment.

Employees responsible for bird care worked on a rotation system so that each person spent an equal amount of time working in the different systems. Labor h for tasks in each system were

Table 1. Replicate and hen numbers for each housing system.

System	# Replicates	Cages/replicate	Hens/Replicate	Total hens housed
Conventional cage	24	2	24	576
Modified cage	36	1	31	1,126
Cage-free	10	1	60	600
Free-range	10	1	60	600

recorded daily from 33 to 89 wk of age and averaged on a 28-day period basis. Egg production, mortality, and feed distribution by weight were monitored daily with feed weigh backs taken at the close of each 28-day period. The cage-free and free-range systems required additional tasks such as litter management and rotating birds between paddocks, respectively.

Free-Range Layout

The free-range layout consisted of 3 free-range houses with 4 replicate pens within each house. Each pen was associated with two 5.57 m^2 (60 ft^2) paddocks, which the hens were allowed continuous access to, alternating between the 2 paddocks every 28 d to encourage pasture regrowth. Paddocks were mowed as needed. A shaded bare dirt veranda area, 13.98 m^2 (150.48 ft^2), was available immediately outside of each replicate pen and opened into the associated paddocks. The indoor portion of the pen contained a slatted floor, 1 tube feeder, 8 nipple drinkers, and 12 nest boxes (at a ratio of 5 hens/nest). Another tube feeder was also available in the paddock area as well as 8 nipples drinkers in the veranda. At the beginning of the lay phase, each replicate contained 60 hens for 0.12 m^2 (1.3 ft^2) per bird.

Cage-Free Layout

The cage-free hens were housed in a high-rise, environmentally controlled facility containing 36 pens that offered a combination of slat and litter flooring. Each pen was 7.43 m^2 (80 ft^2) with 3.53 m^2 (38 ft^2) being slat flooring and 3.90 m^2 (42 ft^2) being litter. Total perch space was 16.3 cm (6.4 in) per bird in accordance with United Egg Producers guidelines (UEP, 2017a,b). Brooding occurred on the litter portion of the floor, and birds were allowed access to the slat section once they were older. Each pen contained 2 tube feeders, 9 nipple drinkers, and 12 nest holes (at a ratio of 5 hens/nest). Each replicate pen contained 60 hens for a stocking density at lay of 0.12 m^2 (1.3 ft^2)/hen.

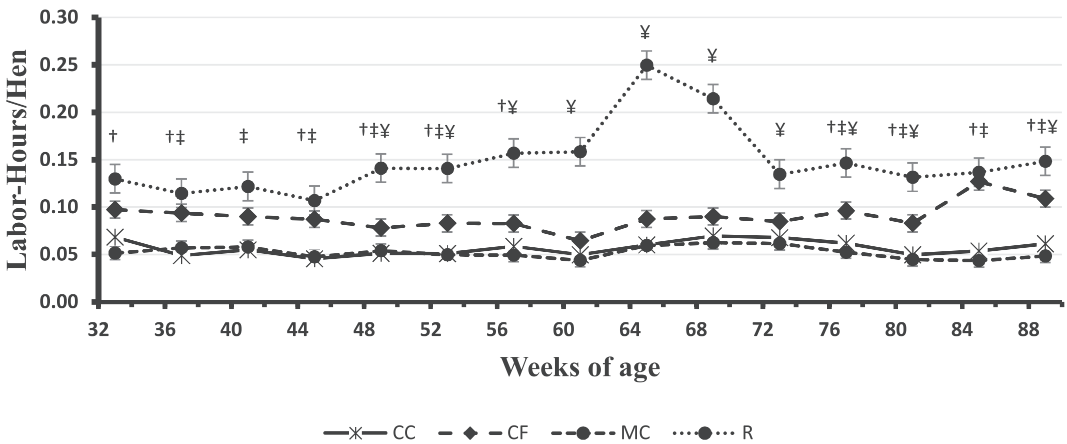

Figure 1. Labor h per hen housed – Effect of conventional cage (**CC**), cage-free (**CF**), modified cage (**MC**), and free-range (**R**) production systems from 33 to 89 wk of age (mean ± SEM). Birds in CC and MC groups did not differ significantly. The R hens required significantly ($P < 0.0001$) more labor hours compared with CC or MC hens throughout the study. Data are presented as mean ± SEM (for n = see Table 1). Symbols indicate significant ($P < 0.0001$) differences between CF hens and CC (‡), MC (†), or R (¥) hens.

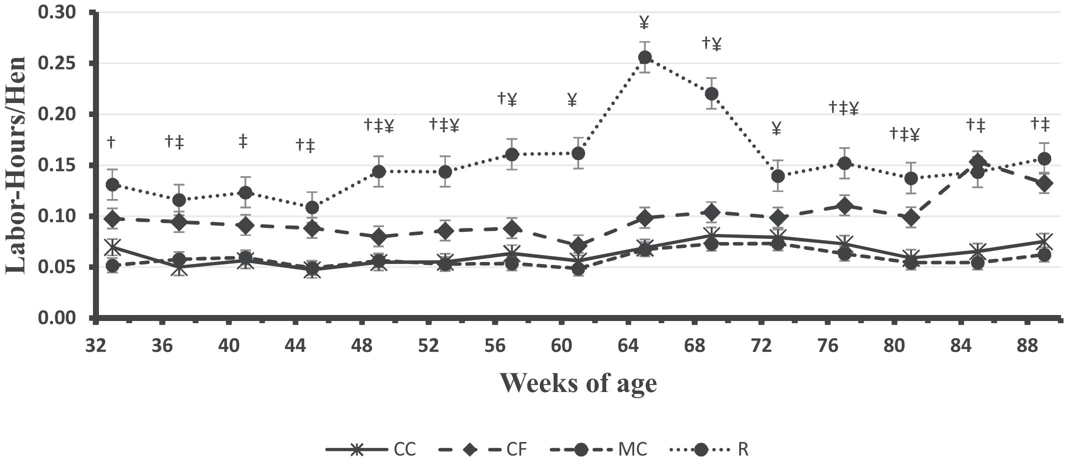

Figure 2. Labor h per hen survived – Effect of conventional cage (**CC**), cage-free (**CF**), modified cage (**MC**), and free-range (**R**) production systems from 33 to 89 wk of age. Birds in CC and MC groups were not shown to differ significantly. The R hens required significantly ($P < 0.0001$) more labor hours compared with CC or MC hens throughout the study. Data are presented as mean ± SEM (for n = see Table 1). Symbols indicate significant ($P < 0.0001$) differences between CF hens and CC (‡), MC (†), or R (¥) hens.

Modified Cage and Conventional Cage Layout

Both cage system houses consisted of 4 banks with 3 tiers of cages within a windowless environmentally controlled building. Each side of a bank was designated as a row, and each row was divided into replicates. In the modified cage house, there were 9 replicate cages per row with each cage being 1.61 m^2 (17.33 ft^2). Each modified cage was either enriched with a nest tent, perches, and scratch pad or barren cage of the same dimensions but without amenities. Hens were divided equally between the 2 cage types, and the tasks required for both types of cages were the same. Each replicate contained 6 drinking nipples as well as 0.66 m (26 in) of feeder trough space. Stocking density at the beginning of lay was 0.05 m^2 (80 in^2) per hen with 31 hens being placed in each replicate cage. Replicate cages were equipped with feed hoppers to supply and monitor feed consumption for individual replicate data; the hoppers were attached to an automatic feed distribution system, although feed was weighed and added manually as needed. Manure was removed from both houses using a belt system under each row that emptied into an auger.

In the conventional cage house, each replicate contained 4 cages that were 0.31 m^2 (3.3 ft^2) in size with a total of 6 replicates occurring on each row. Each individual cage contained on average 2.5 drinking nipples (for a total of 10 per replicate), and the hens had access to 0.61 m (24 in) of feeder trough space. At the start of lay, hens were housed at a stocking density of 0.05 m^2 (80 in^2) per hen with 24 hens being placed in each replicate.

Table 2. Effect of production system on total labor h per hen housed and hen survived from 33 to 89 wk of age.

Production system	Hen housed (h/hen)	Hen survived (h/hen)
Conventional cage	0.057^{C}	0.064^{C}
Modified cage	0.051^{C}	0.058^{C}
Cage-free	0.090^{B}	0.100^{B}
Free range	0.149^{A}	0.150^{A}
SEM	0.003	0.005
P-value	0.001	0.0001

$^{A\text{-}C}$Means in a column that do not possess a common superscript differ significantly ($P \leq 0.001$).

Collection of Man H Data

The electronic data collection system was activated at 33 wk of the production cycle. Time spent within each production system was recorded daily to the quarter h as tasks were completed and converted to total man h for each of the production periods. Daily tasks for which time was recorded consisted of daily collection of production data, mortality, and

Table 3. Effect of production system on labor costs (USD$) per hen and per dozen eggs produced as calculated on hen housed and hen survived bases from 33 to 89 wk of age.

Production system	Hen housed ($/hen)	Hen survived ($/hen)	Hen housed ($/doz)	Hen survived ($/doz)
Conventional cage	0.41^{C}	0.46^{C}	0.27^{C}	0.27^{C}
Modified cage	0.37^{C}	0.42^{C}	0.24^{C}	0.24^{C}
Cage-free	0.65^{B}	0.72^{B}	0.39^{B}	0.40^{B}
Free range	1.08^{A}	1.09^{A}	0.56^{A}	0.57^{A}
SEM	0.02	0.03	0.02	0.02
P-value	0.0001	0.0001	0.0001	0.0001

$^{A\text{-}C}$Means in a column that don't possess a common superscript differ significantly ($P \leq 0.001$).

feed allocation as well as monthly (28-day) feed weigh backs. General system management unrelated to bird care, such as paddock rotation, manure removal, and routine maintenance, were also documented per system. The time spent by each technician in the different management systems were entered into a spreadsheet on Google Drive (G Suite, Google Cloud, Mountain View, CA) using Apple iPads (Apple Inc, Cupertino, CA) as the technicians were inside the houses. Each technician was linked to an individual Google Drive account that was able to track their activities, maintain a history of their data captures, and was compliant with Good Laboratory Practices (USFDA, 2014). Total labor h were compiled per management system for each 4-wk period, along with hen mortality and egg production.

Statistical Analysis

Data were transformed to reflect labor h per bird in terms of the number of hens originally housed and by hens surviving at the end of each period, using the formula from the study by Anderson (2014). Transformed data were analyzed using the Fit Model platform in JMP Pro13 (Sas Int. Inc., 2017) with the effect of system being nested within week of age. Values are presented as least square means, and differences were considered significant at a level of $P < 0.05$ using Tukey's honestly significant difference test. Egg production in terms of eggs per hen (hens originally housed and hens surviving) were also analyzed to compare housing systems. Data were transformed to labor cost per dozen eggs to be evaluated between the housing systems by dividing the labor h by dozen eggs produced per hen (hens originally housed and hens surviving) and then multiplying by an hourly rate for the cost of labor. The federal minimum wage rate of $7.25 per hour was used as a standard labor cost (U.S. Department of Labor, 2019).

RESULTS AND DISCUSSION

Effect of Management System on Labor H Required

The influence of production system on labor h/hh is shown in Figure 1. The conventional and modified cage systems remained similar throughout the trial in terms of labor h/hh and remained within the range of 0.04 to 0.06 h/hh for the duration trial. Free-range birds required more h/hh overall, with time commitment peaking at approximately 65 wk of age ($P < 0.0001$). The rise in labor h for the free-range system during this time coincided with summer and represents increased time required to manage the outdoor paddock during the warmer months. Labor h for the cage-free system fluctuated between 0.6 and 0.13 h/hh throughout the trial and as such, provided an intermediate value between the free-range and cage systems ($P < 0.001$). However, cage-free labor was observed to increase toward the end of the trial owing to increased labor required for litter management within the system. The systems followed a similar trend for labor h per hen survived (**h/hs**), as observed in Figure 2. The free-range system remained the most time demanding in terms of h/hs while both cage systems remained similarly low and the cage-free system fluctuated as an intermediate ($P < 0.001$). Previous work under similar conditions noted increased labor h/hs, owing to a rise in cumulative mortality without an associated decrease in time required for bird management (Anderson, 2014). These data were supported in the present study as time required to manage the different systems remained similar to that of hen housed, despite the decrease in bird numbers.

As shown in Table 2, the overall labor h/hs were higher than those per hen housed, with labor h increasing as the production systems became less intensive ($P < 0.001$). These results support a previous finding with brown egg layers in cage, cage-free, and free-range systems and emphasize the improved efficiency associated with intensive management systems through reduced labor h per hen (Anderson, 2014).

Effect of Management System on Egg Production

To evaluate the economic impact of differences in labor h between the production systems, we calculated the cost of labor h per hen as well as the price per dozen eggs produced (Table 3). Overall, the cost of labor per bird was greater than the price per dozen eggs produced across all management systems for both hen housed and hen survived. Labor costs per hen survived were greater than that for hen housed as labor h per bird was not shown to decrease with cumulative mortality. As the free-range system was earlier demonstrated to be the most labor intensive, it was likewise the most expensive in terms of labor cost per hen housed and hen survived ($P < 0.0001$). Although the price per dozen eggs produced per hen was also highest in the free-range system ($P < 0.0001$), for both hen housed and hen survived, the amount was less than half of the cost of the labor required. Both cage systems represented the lowest labor cost as well as the lowest price per dozen eggs produced; however, the difference between the 2 amounts was lesser than that of the free-range. The labor cost per hen presented here solely represents that of labor input required to manage each production system and does not consider other expenses such as feed or utilities. Although eggs from extensive production systems tend to fetch premium prices, market demands and fluctuations in input costs do not guarantee a profit for the producer (Chang et al., 2010). Producers should consider the economic impact of the increased labor demands associated with extensive management systems for laying hens when evaluating the profitability of converting to alternative production systems.

CONCLUSIONS AND APPLICATIONS

1. Labor input was inversely related to bird stocking density, with intensive cage systems requiring the lowest labor h per hen and extensive range systems the highest.
2. Labor h were increased for hens survived compared with hens housed, as time commitment remained stable despite cumulative mortality. However, the trend in labor required between the systems remained similar for both hens housed and hens survived.
3. The cost of labor per hen was greater than the price per dozen eggs produced per hen in all productions systems. The free-range system demonstrated the highest labor cost and highest price per dozen eggs, while the cage systems represented the lowest for both values.

DISCLOSURES

The authors declare no conflict of interest.

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